HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F70LA

Life Insurance Mathematics A

Semester 1 2021/22

Duration: Two Hours

Please answer all 6 questions.

Total marks: 60

Electronic calculators may be used, provided that they conform to University regulations.

All work must be shown to receive full credit for correct answers.

Throughout this exam, define X as "the last 5 digits of your Heriot-Watt student ID number".

Also, define Y% as [(last digit+1)/10] %.

Also, define Z as "the last 2 digits"

Also, define K as 60 + Z/7, rounded to the nearest integer.

For instance, if your number is H00123455,

then X = 23455

and Y% =0.6% [not 6%]

and Z = 55

and K = 68

Question 1

The following distribution function is used to model the future lifetime of a newborn:

$$F_0(t) = \left(\frac{t}{120}\right)^2$$

(a)Write the survivorship function for a newborn and state acceptable values of t. Describe the upper limit of t using actuarial terminology and common language easily understood by any university student.

[2 marks]

(b) Clearly illustrate that this distribution satisfies the requirements for a survivorship function of a newborn. [3 marks]

(c) Write an expression for ${}_tp_x$ based on the distribution function provided for a newborn. State acceptable values for x+t. Explain why this is also a survivorship function without repeating part (b).

[3 marks]

(d) Calculate $E(T_{80})$ for Ted. Do not be concerned with the fact that the expected age at death is rather high.

[3 marks]

(e) Calculate $E(T_{80})$ for Sam by assuming he skydives for the next 3 years. The only actuarial adjustment for Sam is that he has a constant force of mortality for the next 3 years, $\mu = .08$. If alive at 83, Sam will quit skydiving and his updated complete expectation of life will be 19.62 years.

[4 marks]

(f) Does $E(T_{80})$ for Sam seem appropriate compared to $E(T_{80})$ for Ted? Explain why it seems appropriate or explain any adjustment you would make to μ during the skydiving years.

[2 marks]

[Total 17 marks]

PLEASE TURN OVER

Question 2

A 20-year annual premium endowment insurance is sold to a life aged 40. Annual gross premiums of £2,000 are payable for the term of 20 years. The initial sum insured is £40,000 and the benefit is paid at the end of the year of death or at maturity. Compound reversionary bonuses of 3% per year are declared **after** death benefits are paid each year, just **before** any cash flow occurs for those still alive.

For all your calculations, use the following basis:

Mortality: AM92 ultimate

Interest: 6% per year

Expenses: 2% of each premium and £200 when the benefit is paid

Future bonus additions: 1.92308% per year compound

(a) Calculate the expected present value of the benefit, rounded to the nearest integer. [5 marks]

(b) Calculate the gross premium policy value just before the eighth premium is paid (immediately after the bonus declaration). [3 marks]

[Total 8 marks]

PLEASE TURN OVER

In questions 3 and 4 you should use this ultimate life table.

x	q_x	l_x	d_x	μ_x	e_x
60	0.008016	90975	729	0.007675	22.35
61	0.008773	90246	792	0.008430	21.52
62	0.009492	89454	849	0.009174	20.71
63	0.010259	88605	909	0.009904	19.90
64	0.011227	87696	984	0.010766	19.10
65	0.012404	86712	1076	0.011861	18.32
66	0.013669	85636	1171	0.013109	17.54
67	0.015008	84465	1267	0.014416	16.78
68	0.016595	83198	1381	0.015877	16.02
69	0.018538	81817	1517	0.017674	15.29
70	0.020691	80300	1661	0.019789	14.56
71	0.022860	78639	1798	0.022010	13.86
72	0.025081	76841	1927	0.024243	13.17
73	0.027479	74914	2059	0.026590	12.50
74	0.030193	72855	2199	0.029194	11.84
75	0.033362	70656	2357	0.032210	11.19
76	0.037052	68299	2531	0.035757	10.56
77	0.041194	65768	2709	0.039830	9.95
78	0.045786	63059	2888	0.044363	9.35
79	0.051112	60171	3075	0.049514	8.78
80	0.057401	57096	3277	0.055621	8.22
81	0.064550	53819	3474	0.062774	7.69
82	0.072363	50345	3643	0.070787	7.19
83	0.080889	46702	3778	0.079576	6.71
84	0.090289	42924	3876	0.089302	6.26
85	0.100675	39048	3931	0.100159	5.83

PLEASE TURN OVER

Question 3

(a) Suppose that someone is aged K at 1 January 2021. Using the life table above, what is the probability that the person will be alive on 1 January 2031? Use 6 decimal places.

[1 mark]

(b) Estimate the probability that someone aged K + 0.5 will die in the next 9 months…

(i) …using Uniform Distribution of Deaths, UDD, over ages (K, K+1)

[2 marks]

(ii) …assuming a constant force of mortality over ages (K, K+1)

[4 marks]

(iii) Write down the standard actuarial notation for this probability.

[1 mark]

(iv) Discuss whether it would be appropriate to calculate this probability using only the figure for age K in the column headed μ_x.

[2 marks]

(c) Explain the purpose of using UDD or constant force of mortality assumption with this type of life table. [2 marks]

(d) A student proposes to assume a constant force of mortality over ages (K, K+3).

(i) Calculate the probability that a person aged K+.75 is alive 6 months later.

[3 marks]

(ii) Don't restrict your thoughts to this life table or the ages presented in this life table. Thinking about the variety of life tables and survival distributions you've been exposed to for humans, give an age range where this assumption could be appropriate **and** would not be appropriate. Explain why the assumption is appropriate or not.

[3 marks]

[Total 18 marks]

PLEASE TURN OVER

Question 4

(a) A man aged K takes out a policy that provides a death benefit of X, payable at the end of the year of death, if death occurs after age 82. $A_{85} = 0.8$ and $A_{86} = 0.813193$

The policy is paid for by a single premium at age K.

Calculate the premium.

Basis

Mortality: life table above.

Interest rate: 4%

[Total 6 marks]

PLEASE TURN OVER

Question 5

Calculate $(IA)_{29}$ given that:

Nominal interest rate compounded quarterly: 4% + Y%
$(IA)_{28} = 6$
q_{28} = Y% (written in decimal format)
$\ddot{a}_{28} = 17$

[Total 5 marks]

PLEASE TURN OVER

Question 6

You are given this life table:

X	$l_{[x]}$	$l_{[x]+1}$	l_{x+2}
50	9706	9687	9661
51	9681	9660	9630
52	9653	9629	9596

(a) Write down the standard actuarial notation for an annually increasing life contingent annuity due, that has a last potential payment at age 54, issued to a person that met selection criteria at age 51

[1 mark]

(b) How many years is the selection period for this table?

[1 mark]

(c) Calculate the value of the symbol in part (a) using an effective interest rate of 2% + Y% [4 marks]

[Total 6 marks]

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